

Improving the Efficiency of an EEG-based Brain Computer Interface using Filter Bank Common Spatial Pattern

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Abstract—Brain Computer Interface (BCI) systems are popular due to their ability to improve the quality of life of disabled people. They are proper tools to translate the movement intentions of human into control's signals for out devices. We use Filter Bank Common Spatial Pattern (FBCSP) which leads to extracting discriminative spatial patterns of Electroencephalogram (EEG) signals. In the FBCSP algorithm, the EEG signals are decomposed into multiple frequency bands using a filter bank. In order to exploit the time-varying feature of EEG signals and extract the unknown features, we partitioned the EEG signals into a number of time windows. Then, CSP features are extracted from all time windows of all frequency bands. Finally, feature selection and classification has been done using Mutual Information (MI) algorithm and Naive Bayesian classifier, respectively. We used EEG dataset BCI Competition IV dataset 2a to evaluate the proposed method. This dataset is a four-class two-session motor imagery dataset. The experimental results show that the mean kappa value has improved to 0.60.

Keywords—brain computer interface; motor imagery; common spatial pattern; temporal segmentation; naive bayesian classifier

I. INTRODUCTION

The Brain Computer Interface (BCI) has been widely researched to decoding the intention of a person [1, 2]. According to the type of motor imagery, different patterns of measurable Electroencephalogram (EEG) signals are created in the brain. Computers detect the motor imagery patterns in EEG-based BCI and translate them into control commands for out devices. The aim of brain computer interface systems is providing the communication and motion ability for disabled people who suffer from motor disabilities [3]. They create the interaction with external environment for humans, without any involvement of peripheral nerves and muscles, using control commands generated from brain signals [4]. A BCI enables humans to control a machine (e.g. wheelchair), or control a computer application (such as cursor movement) using only the

brain [5]. There are different EEG signals, such as P300 event-related potentials [6, 7], mu and beta rhythms, visual evoked potentials [8] and slow cortical potentials [9] that are used in BCI systems.

In BCI studies, the motor imagery is the most commonly used mental control strategy, because it can be associated with an attenuation or increase of localized neural rhythmic signals called Event Related Desynchronization/Synchronization (ERD/ERS) [10, 11] that are variations in the amplitude of sensorimotor rhythms [11]. These variations occur in the corresponding signal power of μ (8-12 Hz) and β (13-30 Hz) rhythms in the sensorimotor area [12]. ERD/ERS caused by motor imagery are the most commonly used patterns in BCI systems.

Common Spatial Patterns (CSP) algorithm is one of the popular algorithms in the field of motor imagery-based BCI. The CSP is a decomposition method that given two classes of motor imagery finds optimal spatial filters for maximizing the variance of signals between two classes. Thus it calculates a set of spatial filters that jointly diagonalize the covariance matrices corresponding to two classes of data [13]. In the CSP algorithm, the eigenvectors corresponding to the largest and smallest eigenvalues are generally used as the most discriminative filters for the classification. Despite the efficiency and popularity of the CSP algorithm, it is very sensitive to noise and prone to overfitting [14]. To investigate this issue, the different versions of the CSP algorithm such as Sub-band Common Spatial Pattern (SCSP) [15], Filter Bank CSP (FBCSP) [16], Sparse FBCSP (SFBCSP) [17] and Common Spatio-Spectral Pattern (CSSP) [18] have been proposed. In [16], FBCSP is implemented along with Mutual Information based Best Individual Feature (MIBIF) as its feature selection algorithm and Naive Bayesian Parsen Window (NBPW) as its classification method. In the FBCSP algorithm, filter bank performs frequency filtering and artifact removal [19].

In this study, we propose using the filter bank CSP along with time windowing on the EEG signals to improve four-class classification results on dataset 2a of BCI Competition IV. In the proposed method, the EEG signals are band-pass filtered into multiple frequency bands using a filter bank. Then, the CSP algorithm is applied to each time window for feature extraction in each frequency band. The features obtained from all time windows of all frequency bands construct feature vector. In this paper, we employed a feature selection algorithm to select discriminative features. Finally, Naïve Bayesian classifier is used for four-class classification. The experimental results are compared to three of the best competitors and other studies done on the mentioned dataset.

The remaining sections of the paper are organized as follows. In Section II, we describe the four-class motor imagery based BCI experiment. Then, the proposed method is presented in Section III. Feature extraction and classification methods are briefly explained in this section too. Finally, the experimental results are presented and discussed in Sections IV and V, respectively.

II. DATASET 2A OF BCI COMPETITION IV DESCRIPTION

We used dataset 2a of BCI Competition IV that was presented by Graz University [20]. This dataset comprises EEG data from 9 healthy subjects performing four classes of motor imagery: movement of the right hand, left hand, feet, and tongue. The dataset is recorded in two sessions on different days for each subject. Dataset one of sessions used for training and the other for evaluation. Each of the training and evaluation sessions consists of six runs separated using short breaks and each of run contains 48 trials (12 trials for each class). Indeed, each session consisted of 288 trials (72 trials per class). The signals were recorded from the 22 EEG channels in accordance with the international 10-20 system and 3 monopolar EOG channels. They were sampled with 250 Hz sampling rate and band-pass filtered in the frequency range of 0.5-100 Hz. An additional 50 Hz notch filter has been utilized to suppress power line noise. The EOG channels are used for the subsequent application of artifact removal methods and should not be used for classification. Fig. 1 shows timing scheme of a single trial of motor imagery. During the first two seconds, a fixation cross is displayed on the black screen and then a cue appears during the next second. From $t=3s$ to $t=6s$ the subjects are asked to carry out the motor imagery task. The montage of EEG channels is shown in Fig. 2.

III. METHOD AND MATERIAL

Fig. 3 shows the architecture of the proposed methodology. In this study, the motor imagery EEG signals processing consists of four stages: multiple band-pass filtering using Butterworth filters of order 5, feature extraction by CSP algorithm, selecting the effective features using feature selection algorithm, and classification by Naïve Bayesian

classifier. We considered the time segment of $t=2s$ to $t=8s$ of each trial for processing. This interval is segmented into one-second segments that have 50% overlap. By breaking the EEG signal into a number of time windows with the same length, noise for each time window only affects the features extracted from that window not all features [13]. This reduces the effect of noise on the extracted feature of the EEG signals. The high percentage of overlap of windows creates large dimension feature vectors that cause classifier can't discriminate them well [13]. That is why we used the overlap of 50%. For each trial, the extracted features from all one-second segments of all sub-bands are placed in a feature vector. Then, we obtained mutual information between features and their labels and sort them in descending order. The number of selected features has been changed to get the highest mean kappa value for 9 subjects. The selected features are used to train a Naïve Bayesian classifier. We used only the EEG channels for signal processing. The following has been explained each of four stages listed above.

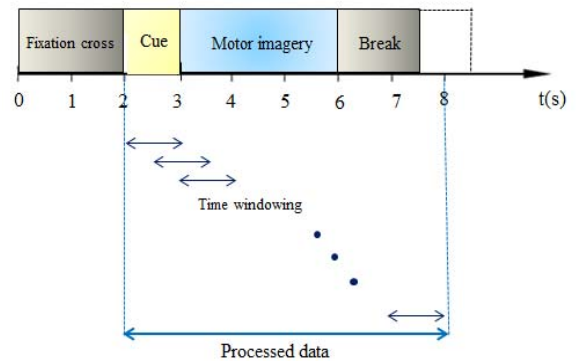


Fig. 1. Timing scheme of a single trial of the motor imagery EEG dataset. The processed temporal segment of each trial and sliding time window have been shown that are used for training and test.

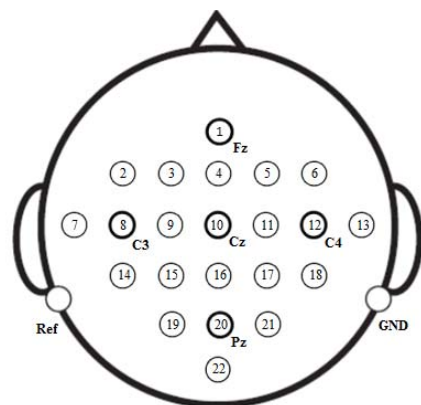


Fig. 2. Electrode montage of the EEG channels.

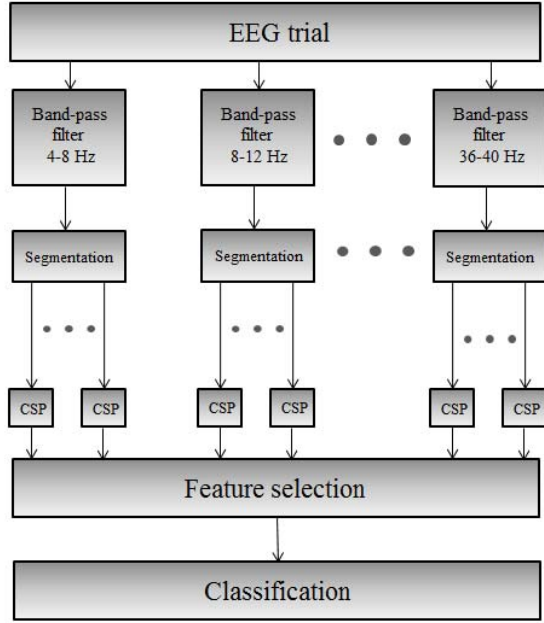


Fig. 3. The architecture of the processing method.

A. Feature Extraction and Selection

1) Band-pass filter

In the first stage, we use filter bank instead of using a wide band filter. Therefore, signals are filtered using 9 sub-bands: 4-8 Hz, 8-12 Hz, ..., 36-40 Hz [21]. Nine Butterworth filters are used to filter the signals in each sub-band.

2) Spatial filtering using common spatial patterns

At second stage, band-pass signals spatially are filtered using CSP [22]. The CSP algorithm is used for spatial filtering the EEG signals into a low-dimensional space by their linearly transformation [23]. Its purpose is to minimize the variance within two classes and maximize the variance between two classes. Spatial filters obtained using the CSP algorithm are used for detecting ERD/ERS [21, 22]. The limitation of the CSP algorithm is possibility of discrimination only two classes [24].

A trial of motor imagery tasks is expressed as $\mathbf{X} \in \mathbb{R}^{N \times T}$, where N , T are the number of channels and the number of samples respectively. The obtained projection matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$ by the CSP algorithm projects trial $\mathbf{X}$ to the brain source signals $\mathbf{S} \in \mathbb{R}^{N \times T}$, as follows:

$$\mathbf{S} = \mathbf{W}\mathbf{X} \quad (1)$$

where the rows of the projection matrix $\mathbf{W}$ are spatial filters. A number of spatial filters of matrix $\mathbf{W}$ are relevant for classification [23]. Thus, after finding the matrix $\mathbf{S}$, the rows

of $\mathbf{S}$ that are corresponding to the first and last m eigenvalues sorted in descending order are selected. Feature related to each selected row of matrix $\mathbf{S}$ obtain as follows:

$$f_q = \log \left(\frac{\text{var}(\mathbf{s}_q)}{\sum_{i=1}^{2m} \text{var}(\mathbf{s}_i)} \right), q \in \{1, \dots, 2m\} \quad (2)$$

where $\text{var}(\mathbf{s}_q)$ is the variance of the q -th row of selected rows from the matrix $\mathbf{S}$.

We obtain Spatial filters using the One-Versus-Rest (OVR) CSP algorithm for a four-class motor imagery-based BCI [25]. Indeed, the CSP algorithm is applied 4 times, and each time the best two spatial filters are selected. Therefore, we obtain 8 features for four-class classification using repetition of the CSP algorithm for each class. Finally, we gather the features corresponding to each trial of each class within a feature vector from all time windows of all 9 sub-bands.

3) Feature selection

In the third stage, a feature selection algorithm is used to select the most discriminative features. In this paper, we used Mutual Information (MI) algorithm [26] for feature selection. The mutual information is computed from each feature with its class label and sorted in descending order. We used the 117 first features which have the higher mutual information for training a Naïve Bayesian classifier.

B. Classification

1) Naïve Bayesian classifier

In this stage, the motor imagery trials are labelled using a classifier. We employed the Naive Bayesian classifier for classification. A Naive Bayesian classifier is a probabilistic classifier that has strong independence assumptions [27]. This classifier assigns a trial to the class with maximum probability. Table I presents the comparative analyses of the classification results between the proposed method in this paper and past studies.

IV. RESULT

In this section, the proposed method is evaluated on publicly available BCI Competition IV dataset 2a. We utilized the FBCSP along with time windowing on the EEG signals to increase the performance of four-class classification. In this paper, performance is computed in terms of kappa [28]. Cohen's kappa coefficient is used to compare our results with three of the best competitors as well as other studies [29-31]. We applied CSP with $m=1$ that means first row and last row from spatial filters are used for features computation. Then, the

MI algorithm is used to select the most effective features. We observed that the highest mean kappa value occurs at 117th feature of the arranged features in descending order. After the 117 first features, mean kappa value has a descending behavior. The kappa values obtained by proposed approach and the other studies have been shown in Table I. Based on Table I, the proposed method and Kam et al. [31] have the same mean kappa value, but our method has the standard deviation of 0.17, which is less than the standard deviation in [31]. The influence of varying the number of selected features on the mean kappa value is shown in Fig. 4.

V. DISCUSSION AND CONCLUSION

In this paper, the problem of four-class motor imagery classification on EEG dataset BCI Competition IV dataset 2a was studied. We implemented FBCSP along with MI feature selection algorithm and Naive Bayesian classifier. Table I shows that the proposed method has a mean kappa value of 0.60.

In order to improve the motor imagery classification results, we should not limit ourselves only to a section of the

signal processing, but should focus on all sections such as preprocessing, feature extraction, feature selection and classification [32]. In this study, we observe that the appropriate selection of the processed time segment of each trial, length of sliding window, overlap percentage of windows and type of extracted features improve our results comparing to the best three competitors as well as other studies [29-31]. In this paper, parameters such as length of sliding window, percentage overlap of windows and the number of selected spatial filters to features computation are constant for all subjects.

For test data, Kam et al. [31] used a sliding 2-second window, which moves from $t=0s$ to $t=7s$ with 10sample intervals. The advantage of the implemented method compared to Kam et al. [31] method is that the number of windows used in our method is significantly less, which makes our method faster. The important note is that they reported the maximum kappa achieved throughout the time course of the paradigm. In future work, we will try to improve the performance of four-class classification on the above mentioned dataset.

TABLE I. COMPARISON OF KAPPA VALUES BETWEEN IMPLEMENTED METHOD AND OTHER STUDIES ON DATASET 2A OF BCI COMPETITION IV.

Method	Subjects										
	<i>S1</i>	<i>S2</i>	<i>S3</i>	<i>S4</i>	<i>S5</i>	<i>S6</i>	<i>S7</i>	<i>S8</i>	<i>S9</i>	<i>AVG.</i>	<i>S.D.</i>
1 st [21]	0.68	0.42	0.75	0.48	0.40	0.27	0.77	0.75	0.61	0.57	0.18
2 nd	0.69	0.34	0.71	0.44	0.16	0.21	0.66	0.73	0.69	0.51	0.23
3 rd	0.38	0.18	0.48	0.33	0.07	0.14	0.29	0.49	0.44	0.31	0.15
Wang [29]	0.67	0.49	0.77	0.59	0.52	0.31	0.48	0.75	0.65	0.58	0.15
Asencio-Cubero et al. [30]	0.75	0.50	0.74	0.40	0.19	0.41	0.78	0.72	0.78	0.59	0.22
Kam et al. [31]	0.74	0.35	0.76	0.53	0.38	0.31	0.84	0.74	0.74	0.60	0.21
Proposed method	0.68	0.39	0.81	0.54	0.37	0.43	0.73	0.73	0.69	0.60	0.17

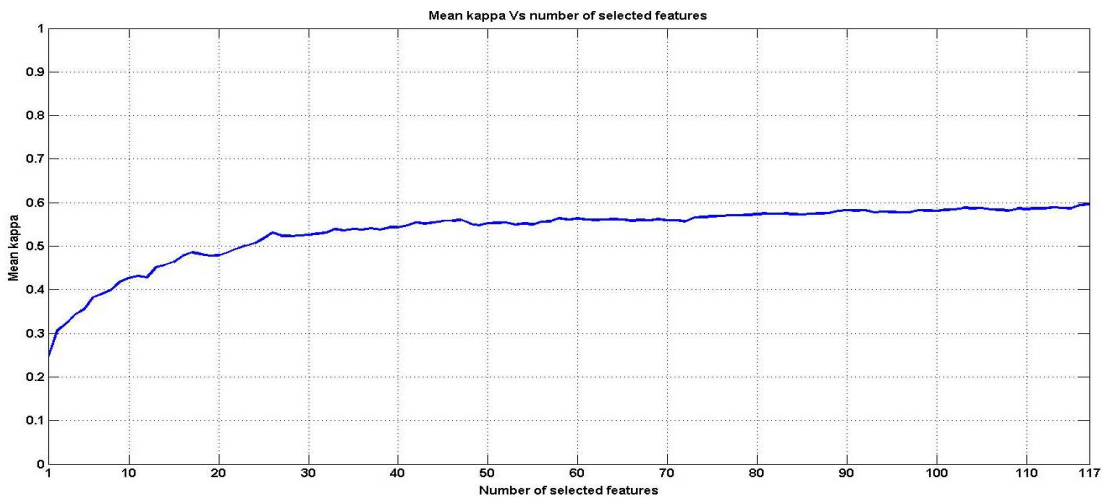


Fig. 4. Variations of the mean kappa value based on the number of selected features for the classifier.

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